

An Experiment in Green Living

Prairie Crossing, a community outside of Chicago, offers families energy-efficient, green living.

by Elka Karl

By spending just an hour on a Metra commuter train, residents can efficiently travel from their home in the Prairie Crossing project to their jobs in downtown Chicago. And when they return home in the evening, these same residents can indulge in just one of the many amenities offered by their residential green community. Perhaps they will pick their children up from the LEED-certified community school, or swing by the farm to pick up some eggs, or even paddle a canoe over the surface of Lake Aldo Leopold, named after the great Wisconsin conservationist.

Prairie Crossing, located in Grayslake, Illinois, is an experiment in green living that was designed with responsible development ethics, good mass transport opportunities, and land preservation in mind. The land on which Prairie Crossing was built was purchased in 1987 by a group of neighbors who wanted to preserve open space and agricultural land. They formed a company with the goal of developing the 677 acres responsibly. George and Victoria Post Ranney, a husband-and-wife team, have guided the development of Prairie Crossing since its beginning. All 359 of the single-family homes have been sold, and the first residents are moving into 36 condos in the latest phase. Given the success and critical acclaim the project has received, it's safe to say that Prairie Crossing is a continuing success.

From Beans to Buildings

The first homeowners moved to Prairie Crossing ten years ago, and the new-construction single-family homes have all been sold, although some are now available for resale by homeowners. The single-family homes of Prairie Crossing are well known for excellent design and architecture. Accomplished architects were asked to look at older homes in nearby towns and on farms, and to develop homes with new designs based on midwestern architectural traditions.

While these homes may look old, their technology is anything but. The homes at Prairie Crossing were built at the same cost as most northwestern Chicago homes, but they are 50% more energy efficient than comparable homes in the Chicago area.

The homes were all built to DOE Building America program standards. Taking a whole-house approach to construction, the homes were designed to make maximum use of the interaction between the building envelope and the heating and cooling system. Dave Richmond, who was employed as director of construction on the Prairie Crossing project, was instrumental in bringing building science on board. "We were able to get into the [Building America] program at the last minute," says Richmond. "We did three prototype homes and were so impressed with



PRAIRIE CROSSING

The Prairie Crossing Organic Farm was one of the first parts of the community to be established and remains at its heart.

the results that we decided to build all of the homes under the Building America criteria."

Richmond notes that the homes all used a tight envelope with advanced framing. One-inch rigid foam insulation was used on the exterior sheathing, with R-19 fiberglass insulation in the walls. Prairie Crossing also used a dry wall air barrier approach in regards to the vapor retarder. Blown cellulose (R-50) was used in the attic, and low-e argon-filled high-performance windows were used throughout the homes. The mechanical system was right-sized for the building, and a fan cycler is used to regulate ventilation pressure, using the existing duct systems.

Asked if he's seen any ripple effects from the Prairie Crossing project, Richmond says, "It's influenced a lot of people. We see a lot of other developments around the country that have taken some of the ideas, certainly with the design and layout of the homes. Wisconsin is replicating what we've done on a smaller scale."

An Eye on Energy Use

Mike Sands, environmental team leader at Prairie Crossing, says that the community is focusing now on occupant behavior. To that end, Prairie Crossing has developed innovative



green building

The Station Square condominiums uphold the high architectural standards for which Prairie Crossing has become known. The Burnham, the Bennett, and the Wacker Buildings, named to commemorate three great urban planners in Chicago's history, were designed by Worn Jerabek Architects. All three buildings are Energy Star rated.

ice skating, sledding, and cross-country skiing in the winter. The Prairie Crossing Farm Market features delicious locally grown organic produce, honey, and eggs. Residents can also keep their horses near their homes, thanks to a cooperative of homeowners with horses that manages the small stable at Prairie Crossing.

Residents with children can apply for admission to the on-site Prairie Crossing Charter School, a public elementary school where learning is focused on the natural environment. The Prairie Crossing Charter School offers elementary education based on an environmental curriculum to children from two local school districts. This new 13,000 ft² charter school, which is LEED certified, takes an ecological,

integrated, experiential approach to learning. The school building was designed to reduce energy use, conserve water, and use recycled-content materials that are manufactured locally.

Every classroom exits directly to the outdoors, which facilitates use of the school garden, which is watered with storm water collected in cisterns. To further reduce storm water runoff, the school also uses porous paving surfaces and vegetated swale. A geothermal heat pump system is used to heat and cool the building, and interior spaces have occupancy sensors to help improve lighting efficiency. The flooring includes renewable natural materials.

ways to encourage efficient energy use. For instance, it is offering a \$1,000 cash prize to the family whose home is the most energy efficient in 2007. The contest is open to all Prairie Crossing single-family residences. The goal is to achieve the lowest electric and natural gas consumption per square foot per family member throughout the 2007 calendar year. Residents can use an easy-to-navigate Excel spreadsheet, downloadable from the Prairie Crossing Web site, to plug in their energy use each month. The winner will be announced (and the check presented) at the 2008 Prairie Crossing Annual Homeowners Association membership meeting.

Prairie Crossing also encourages energy efficiency by offering residents information on energy-efficient home practices, such as turning down thermostats in winter and up in summer and switching out incandescent bulbs with CFLs, and by encouraging the use of public transit, walking, and native landscaping throughout the community.

A Sense of Place

Life at Prairie Crossing is enriched by many community events, such as gatherings at the Byron Colby Barn, gazebo concerts, swimming, and cookouts at the beach on Lake Aldo Leopold in the summer, and

green building

Corridors between wings are separated from breezeway doors. This allows heated or cool air to remain in a designated area of the building. Skylights and open-air ceilings let in natural light to illuminate the corridors. They are also used as teaching tools; by teaching students about energy-efficient and green building, students are given an early lesson in building science.

In the belief that community and conservation can go hand in hand, the trails and gardens of Prairie Crossing are designed to be places where people can meet to enjoy and care for the land. The Homeowners Association has taken responsibility for these and other community amenities. Volunteer stewardship activities are organized by the Liberty Prairie Conservancy, which conducts environmental programs throughout the Liberty Prairie Reserve.

Because Prairie Crossing is located within easy walking distance of two Metra stations, it encourages its residents to take public transit. It is often cited as a transit-oriented development, and many residents of Prairie Crossing chose to live here partly for that reason.

Two Metra commuter lines, with connections to Chicago and O'Hare International Airport, are just a few minutes' walk from Prairie Crossing. Residents can walk to the train and be in downtown Chicago in an hour on the Milwaukee District North Line, or at O'Hare in 35 minutes on the North Central Line. Almost 300 trains a week run on these two lines, offering rail access that is virtually unprecedented for a suburban community.

Farm Life

One of the most treasured buildings in Prairie Crossing is the Byron Colby Barn, which was built in 1885 and restored in 1996 to serve as a community center for Prairie Crossing and the surrounding area. Hundreds of community events take place here each year, including homeowners meetings,



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(top) Two Metra commuter lines, with connections to Chicago and O'Hare International Airport, are just a few minutes' walk from Prairie Crossing. Residents can walk to the train and be in downtown Chicago in an hour on the Milwaukee District North Line, or at O'Hare in 35 minutes on the North Central Line.

(bottom) Prairie Crossing's new 13,000 ft² charter school, which is LEED certified, takes an ecological, integrated, experiential approach to learning.

concerts, lectures, school assemblies, workshops, and conferences. The Barn serves as a polling place and is the base for the annual Prairie Pedal, a popular family bike event run by the Liberty Prairie Conservancy.

But the idea of farm life isn't strictly symbolic at Prairie Crossing. The Prairie Crossing Organic Farm was one of the first parts of the community to be established and remains at its heart. Based on an old farmstead with several houses, barns, and outbuildings, the farm consists of nearly 90 acres and grows healthy, certified-organic produce.

The largest part of the Prairie Crossing Organic Farm is Sandhill Organics, which is run by farmers Matt and Peg Sheaffer. Along with their son Avery, Matt and Peg came to the farm from East Troy, Wisconsin,

where they had farmed organically for several years. Their crops include a wide variety of vegetables throughout the season, as well as berries and cut flowers, which they sell at the Prairie Crossing Farmer's Market. Boxes of Sandhill Organics produce are also distributed once a week during the growing season to members of a community-supported agriculture (CSA) group. CSA members buy shares in the farm in the spring, when there is otherwise little farm income. In exchange, they receive fresh vegetables, herbs, and fruit throughout the growing season. They agree to share the fortunes of the farm, including potential weather-related losses.

The Learning Farm, which is part of Prairie Crossing Organic Farm, leads outdoor, experiential, farm-based education programs and grows food

Energy Features

Building Envelope

- **Ceiling/roof:** Truss with R-40 insulation batts, with vented assembly
- **Walls:** 2 x 6 advanced framing, with R-19 batts instead of standard R-13, R-7 insulating sheathing, and interior air-flow retarder
- **Basement:** R-13 full height fiberglass batt insulation
- **Windows:** Double-glazed, low-e, argon-filled
- **Total envelope leakage:** Less than 2.5 square inches per 100 square feet of building envelope area at 10 Pa

Mechanical System

- **Furnace:** 90% efficient condensing gas furnace; single unit, within envelope
- **Hot water tank:** 75% efficient power-vented gas water heater
- **Ductwork:** leakage less than 5% of total rated flow at high speed at 25 Pa; all ductwork within conditioned space
- **Ventilation:** Supply fan ventilation system with distributed exhaust; independently controlled

sustainably. All of the farm work is done by students and volunteers. It includes 3 acres of certified-organic land, which is used to grow crops, as well as several greenhouses and a flock of free-range chickens. The Learning Farm was developed in 2004 as a program of the Prairie Crossing Institute, the main educational nonprofit organization in Prairie Crossing Conservation Community.

The Learning Farm works with 375 students from two local schools, the Prairie Crossing Charter School (grades K–8) and the Adolescent program of the Montessori School of Lake Forest (grades 7–9). Students from both schools engage in lessons and work experience on the farm—lessons that integrate agriculture with science, math, nutrition, social studies, and economics.

Land Ethics

Over 60% of the 677-acre site at Prairie Crossing is protected open land. Ten miles of trails wind through a landscape of farm fields, pastures, lakes and ponds, native prairies, and wetlands. On these trails, residents walk, run, bike, ski, ride horseback, and watch the many species of birds, butterflies, and other wildlife that are attracted to a healthy ecosystem and native landscaping. In addition, Prairie Crossing is linked by regional trails to

the Liberty Prairie Reserve, over 3,200 acres of legally protected land.

The Liberty Prairie Reserve consists of tallgrass prairie, wetlands, oak savannas, rolling farm fields, and forest; it is listed as one of America's Last Chance Landscapes by Scenic America. Home to 14 threatened or endangered species, it also contains three Illinois Nature Preserves—areas designated for the state's highest level of protection because they consist of rare and endangered ecosystems.

Thanks to a site design that filters storm water runoff through the prairies and wetlands, the water in Prairie Crossing's Lake Aldo Leopold is pure enough to swim in. In fact, the water quality is such that the Illinois Department of Natural Resources selected Prairie Crossing as a site for stocking endangered native minnows that are critical to a healthy biodiversity in streams and lakes. Residents gather at the popular community beach and canoe, kayak, and sail on the lake.

New Developments

At the head of the landscaped Station Square, Prairie Crossing's commercial center, stands the Burnham Building, with 20 condominiums on three floors overlooking the square to the south and the Wild Indigo garden to

the north. Flanking Station Square on either side are the Bennett and Wacker Buildings, each with eight spacious condominiums on two floors above first-floor shops and offices. Sites adjacent to these two buildings are reserved for retail. The first of the retail shops have just opened at Station Square. Prairie Croissant Café offers fair trade coffee and delicious sandwiches and salads, some of which are made with organic produce from the farm. Ten Thousand Villages, a fair trade retail store, is part of the nationwide network of Ten Thousand Villages stores selling exquisite hand-crafted clothing and decorative items made by artisans from around the world. The children's bookstore Under the Sycamore Tree and the children's boutique Little Skye both stock thoughtfully chosen items for local families.

The Station Square condominiums have been planned with care. Each of the 15 floor plans has two or three bedrooms; floor area ranges from 1,596 to 2,720 square feet. Prices start at \$329,000. Condominium owners benefit from a maintenance-free lifestyle with luxury features and security, all within the nationally recognized conservation community.

The three condominium buildings uphold the high architectural standards for which Prairie Crossing has become known. The Burnham, the Bennett, and the Wacker Buildings, named to commemorate three great urban planners in Chicago's history, were designed by Worn Jerabek Architects, a Chicago-based firm with a strong background in environmental design. All three buildings have already received the Energy Star rating—making them unique among multiunit buildings in a three-state area. To earn the Energy Star rating, a home must be independently verified to be at least 30% more energy efficient than a comparable home built to the Model Energy Code.

Exterior walls of the three condominium buildings are highly

green building

insulated and use a state-of-the-art vapor barrier to control water vapor movement in walls. Windows are double glazed, low-e, and argon filled. Low-VOC (volatile organic compound) paints, sealants, and adhesives are used throughout, to promote healthy indoor air quality (IAQ). The furnaces are 93% high efficiency with sealed combustion. These furnaces will reduce energy costs and will also promote healthy IAQ. The high-efficiency sealed-combustion water heaters will further reduce energy costs and increase IAQ. To encourage recycling, the trash chutes are designed to switch between trash and recyclable materials at the touch of a button.

“The energy-efficient condos at Prairie Crossing are part of the whole fabric of the community, which is designed for environmental protection and enhancement,” says Victoria Post Ranney, cofounder and president of



Prairie Crossing is a bike- and pedestrian-friendly community.

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Prairie Crossing. And she adds, “At Prairie Crossing, it’s not just green building—you get to enjoy the pleasures of green living.”

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Elka Karl is an associate editor at Home Energy magazine. Prairie Crossing Holding Corporation provided some of the information used in this article.

For more information:

To learn more about Prairie Crossing, go to www.prairiecrossing.com.

To learn more about Building America, go to www.eere.energy.gov/buildings/building_america.